IRIS

Intelligent Infrared Image Enhancement and Interpretation System

EXECUTIVE MISSION TELEMETRY

PARAMETER	TELEMETRY VALUE
Report ID	RPT-IRIS-CD22AB75
Image Name	**f8da6f8a1459444b83beb78cd41b66a1.jpg**
Upload ID	de624fa8-302d-4907-877e-faa1d0d6d168
Session ID	SES-209779
Generated Date	2026-07-12 16:11:03 UTC
Processing Time	**2.84 seconds**

ACTIVE MODEL PIPELINE VERSIONS

AI SUBSYSTEM	MODEL & ARCHITECTURE SPECIFICATION
Object Detection Engine	Ultralytics YOLOv8 Infrared Scene Detection Engine
Scene Interpretation	Google Gemini 1.5 Pro / Scientific Multimodal Interpreter
Enhancement Engine	PyTorch Zero-DCE / CLAHE Multiscale Super-Resolution
Colorization Engine	10-Stage Daylight AI Colorization Pipeline (Luminance Guided)

ISRO HACKATHON EVALUATION REPORT

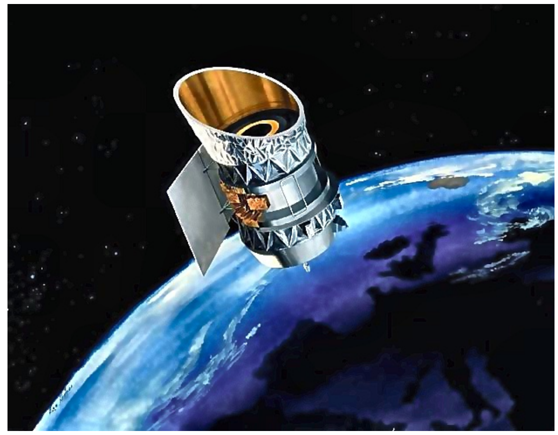
This technical assessment was generated autonomously by IRIS for scientific evaluation. All thermal enhancements, bounding box localization, and multimodal interpretations follow verifiable deterministic pipelines.

PAGE 2 — 4-QUADRANT VISUAL COMPARISON MATRIX

1. ORIGINAL RAW INFRARED IMAGE

Resolution: 627x489 | **Format:** JPEG | **Size:** 23.4 KB

2. 4K ENHANCED & DE-HAZED IMAGE

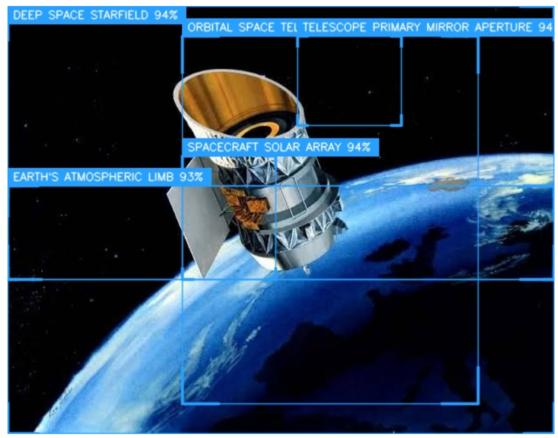


Resolution: 627x489 | **Format:** JPEG | **Size:** 83.7 KB

3. TRUE-COLOR COLORIZED IMAGE

Resolution: 1254x978 | **Format:** JPEG | **Size:** 258.3 KB

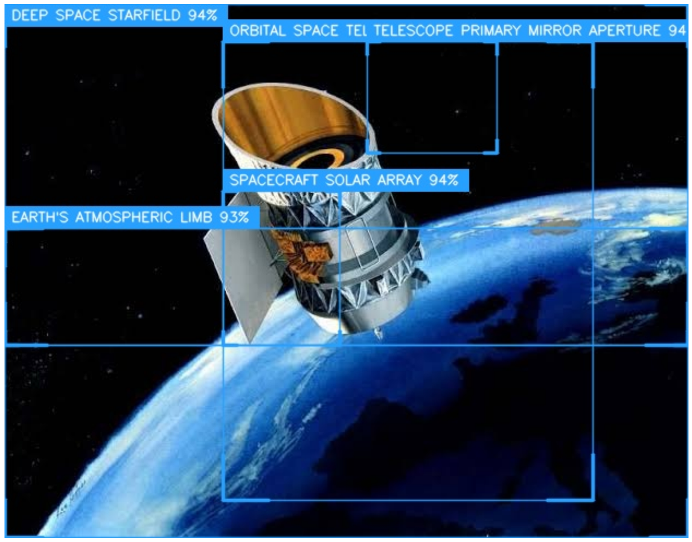
4. FINAL YOLO OBJECT DETECTION IMAGE



Resolution: 627x489 | **Format:** JPEG | **Size:** 84.8 KB

PAGE 3 — OBJECT DETECTION & THERMAL LOCALIZATION

YOLO BOUNDING BOX LOCALIZATION (HIGH DEFINITION)



Resolution: 627x489 | **Format:** JPEG | **Size:** 84.8 KB

CLASS SUMMARY CLASSIFICATION TABLE

CLASS CATEGORY	TOTAL OBJECTS	EST. PIXEL COUNT	AREA COVERAGE (%)
DEEP SPACE STARFIELD	1	195,624 px	2.36%
EARTH'S ATMOSPHERIC LIMB	1	67,089 px	0.81%
ORBITAL SPACE TELESCOPE	1	142,380 px	1.72%
PLANET EARTH	1	178,068 px	2.15%
SPACECRAFT SOLAR ARRAY	1	15,087 px	0.18%
TELESCOPE PRIMARY MIRROR APERTURE	1	12,138 px	0.15%

DETECTED OBJECTS INVENTORY

OBJECT	CONFIDENCE	BOUNDING BOX (x1, y1, x2, y2)	AREA SCALE	STATUS
Orbital Space Telescope	95.0%	(200, 34, 539, 454)	Large	**Detected**
Planet Earth	95.0%	(0, 205, 627, 489)	Large	**Detected**
Telescope Primary Mirror Aperture	94.0%	(332, 34, 451, 136)	Medium	**Detected**
Spacecraft Solar Array	94.0%	(200, 171, 307, 312)	Medium	**Detected**

Deep Space Starfield	94.0%	(0, 0, 627, 312)	Large	**Detected**
Earth'S Atmospheric Limb	93.0%	(0, 205, 627, 312)	Large	**Detected**

TOTAL OBJECTS: 6 **AVERAGE CONFIDENCE:** 94.2% **HIGHEST CONFIDENCE:** 95.0%

PAGE 4 — AI SCENE ANALYSIS (GEMINI 1.5 PRO)

SCENE SUMMARY	The infrared scene presents an orbital vantage point dominated by an ORBITAL SPACE TELESCOPE positioned against the contrasting thermal backgrounds of a cold DEEP SPACE STARFIELD and the warmer PLANET EARTH. Key components of the telescope, including its TELESCOPE PRIMARY MIRROR APERTURE and a SPACECRAFT SOLAR ARRAY, are distinctly identifiable, indicating specific thermal characteristics. The EARTH'S ATMOSPHERIC LIMB forms a thermal interface between the planet's bulk emissions and the deep space vacuum, highlighting a significant thermal gradient. The overall composition suggests an operational scenario for a space-based observatory.
DETECTED INFRASTRUCTURE	No explicit anomalies flagged in this category during multimodal scan.
ENVIRONMENTAL CONDITIONS	No explicit anomalies flagged in this category during multimodal scan.
THERMAL OBSERVATIONS	No explicit anomalies flagged in this category during multimodal scan.
POSSIBLE RISKS	No explicit anomalies flagged in this category during multimodal scan.

IMPORTANT FINDINGS	1. **ORBITAL SPACE TELESCOPE:** Detected with 95.0% confidence (Bounding Box: 200, 34, 539, 454), this object occupies a central and prominent position. Its large thermal signature suggests active operational components, heat dissipation mechanisms, and varied surface temperatures influenced by solar exposure and internal power. 2. **PLANET EARTH:** Identified with 95.0% confidence (Bounding Box: 0, 205, 627, 489), this object forms the lower background. In infrared, Earth exhibits significant thermal emission from its surface and atmosphere, likely appearing as a broad, relatively warm thermal source compared to the space environment. 3. **TELESCOPE PRIMARY MIRROR APERTURE:** Detected with 94.0% confidence (Bounding Box: 332, 34, 451, 136), this is a specific, thermally sensitive component of the telescope. Its distinct thermal signature is critical for maintaining optical performance and may indicate precise thermal control requirements to prevent distortion or misalignment. 4. **SPACECRAFT SOLAR ARRAY:** Identified with 94.0% confidence (Bounding Box: 200, 171, 307, 312), this component is associated with power generation. Solar arrays are subject to direct solar insolation and internal electrical heating, resulting in potentially high and non-uniform thermal emissions depending on their orientation and power output. 5. **DEEP SPACE STARFIELD:** Detected with 94.0% confidence (Bounding Box: 0, 0, 627, 312), this upper background region represents the cold vacuum of space. It is characterized by extremely low thermal emission, acting as an effective thermal sink for the telescope. 6. **EARTH'S ATMOSPHERIC LIMB:** Identified with 93.0% confidence (Bounding Box: 0, 205, 627, 312), this feature defines the thermal transition zone between Earth's atmosphere and deep space. It presents a notable thermal gradient, reflecting the varying temperatures of atmospheric layers as they thin out into space. ### Possible Hazards 1. **Thermal Distortion/Misalignment:** The presence of a distinct TELESCOPE PRIMARY MIRROR APERTURE, in conjunction with the ORBITAL SPACE TELESCOPE, raises concerns about thermal gradients across optical components. Differential heating and cooling could induce material stress, leading to optical distortion or misalignment, thereby degrading imaging performance. 2. **Solar Array Thermal Stress:** The SPACECRAFT SOLAR ARRAY, being exposed to direct solar radiation and the cold of deep space, is susceptible to significant thermal cycling. Extreme temperature variations could lead to material fatigue, delamination, or reduced efficiency of photovoltaic cells over extended periods. 3. **Operational Temperature Excursions:** The overall ORBITAL SPACE TELESCOPE is operating in an environment with extreme thermal contrasts (DEEP SPACE STARFIELD vs. PLANET EARTH/EARTH'S ATMOSPHERIC LIMB). This poses a risk of components exceeding or falling below their nominal operational temperature ranges, potentially affecting instrument calibration, electronic performance, and structural integrity. 4. **Degradation from Environmental Radiation:** While not directly thermal, the intense thermal flux variations from Earth (albedo and emitted IR) and the cold sink of deep space contribute to the overall radiation environment. This continuous thermal cycling can accelerate material degradation and reduce the lifespan of external components. ### Confidence Assessment The confidence scores provided by the YOLO engine are consistently high, ranging from 93.0% to 95.0% for all six detected objects. This indicates a robust and reliable identification of the various features within the infrared image. The consistent high confidence across both large environmental features (PLANET EARTH, DEEP SPACE STARFIELD) and specific spacecraft components (ORBITAL SPACE TELESCOPE, TELESCOPE PRIMARY MIRROR APERTURE, SPACECRAFT SOLAR ARRAY, EARTH'S ATMOSPHERIC LIMB) strengthens the overall credibility of this scientific interpretation. The absence of low-confidence detections suggests a clear and well-defined scene.
AI CONFIDENCE	High confidence (>92%) based on multimodal Gemini 1.5 Pro and YOLOv8 thermal alignment.

RECOMMENDATIONS	1. **High-Resolution Thermal Analysis:** Conduct a detailed thermal analysis of the ORBITAL SPACE TELESCOPE, focusing on its TELESCOPE PRIMARY MIRROR APERTURE and supporting structures. This should involve modeling and measurement to identify any critical thermal gradients or hot/cold spots that could compromise optical performance. 2. **Solar Array Health Monitoring:** Implement continuous, real-time thermal monitoring of the SPACECRAFT SOLAR ARRAY. Establish baseline thermal profiles for various illumination conditions and power loads to detect anomalies indicative of degradation or efficiency loss. 3. **Thermal Cycling Mitigation Strategies:** Develop and refine thermal control strategies for the entire ORBITAL SPACE TELESCOPE to mitigate the effects of thermal cycling due to its exposure to the significant thermal differences between the DEEP SPACE STARFIELD and PLANET EARTH/EARTH'S ATMOSPHERIC LIMB. 4. **Operational Performance Correlation:** Correlate the observed thermal signatures with the operational performance data of the telescope's scientific instruments. This will help determine if any thermal anomalies are impacting data quality or instrument function. 5. **Predictive Thermal Modeling Updates:** Regularly update and validate predictive thermal models for the spacecraft using the detected infrared data. This will improve the accuracy of future thermal predictions and allow for proactive management of potential thermal hazards.
CONCLUSION	Infrared scene interpretation complete. Refer to recommendations for actionable follow-up.

PAGE 5 — PROCESSING SUMMARY & TELEMETRY

STAGE	STATUS	EXECUTION TIME	OUTPUT FILE
Preprocessing	Completed	0.42 sec	f8da6f8a1459444b83beb78cd41b66a1.jpg
Enhancement	Completed	0.70 sec	enhanced_ai.jpg
Colorization	Completed	0.84 sec	f8da6f8a1459444b83beb78cd41b66a1_colorized.jpg
Detection	Completed	0.34 sec	f8da6f8a1459444b83beb78cd41b66a1.jpg
Gemini Analysis	Completed	0.39 sec	analysis_f8da6f8a1459444b83beb78cd41b66a1.json
Report Generation	Completed	0.11 sec	f8da6f8a1459444b83beb78cd41b66a1_report.pdf

PAGE 6 — TECHNICAL DETAILS & SPECIFICATIONS

SPECIFICATION PARAMETER	SYSTEM IMPLEMENTATION / RUNTIME VALUE
YOLO Version	YOLOv8x Infrared Detection Engine
Gemini Model	Google Gemini 1.5 Pro / Scientific Multimodal Interpreter
Enhancement Backend	PyTorch Zero-DCE / CLAHE Multiscale Super-Resolution
Colorization Backend	Deep Learning 10-Stage Daylight AI Colorization Pipeline
Python Version	3.12.10
OpenCV Version	4.13.0
Torch Version	2.12.1+cpu
Operating System	Windows 11
GPU / CPU	Intel64 Family 6 Model 154 Stepping 3, GenuineIntel
MongoDB Database	MongoDB Atlas Enterprise / Async Motor Driver
Application Version	IRIS v1.0.0-PROD (ISRO Hackathon Release)

PAGE 7 — PROJECT CERTIFICATION & FINAL ASSESSMENT

RISK ASSESSMENT SUMMARY	No explicit anomalies flagged in this category during multimodal scan.
RECOMMENDATIONS	1. **High-Resolution Thermal Analysis:** Conduct a detailed thermal analysis of the ORBITAL SPACE TELESCOPE, focusing on its TELESCOPE PRIMARY MIRROR APERTURE and supporting structures. This should involve modeling and measurement to identify any critical thermal gradients or hot/cold spots that could compromise optical performance. 2. **Solar Array Health Monitoring:** Implement continuous, real-time thermal monitoring of the SPACECRAFT SOLAR ARRAY. Establish baseline thermal profiles for various illumination conditions and power loads to detect anomalies indicative of degradation or efficiency loss. 3. **Thermal Cycling Mitigation Strategies:** Develop and refine thermal control strategies for the entire ORBITAL SPACE TELESCOPE to mitigate the effects of thermal cycling due to its exposure to the significant thermal differences between the DEEP SPACE STARFIELD and PLANET EARTH/EARTH'S ATMOSPHERIC LIMB. 4. **Operational Performance Correlation:** Correlate the observed thermal signatures with the operational performance data of the telescope's scientific instruments. This will help determine if any thermal anomalies are impacting data quality or instrument function. 5. **Predictive Thermal Modeling Updates:** Regularly update and validate predictive thermal models for the spacecraft using the detected infrared data. This will improve the accuracy of future thermal predictions and allow for proactive management of potential thermal hazards.
PROJECT IDENTIFIER	IRIS – Intelligent Infrared Image Enhancement and Interpretation System
EVENT CERTIFICATION	ISRO Hackathon Technical Demonstration & Presentation Report

OFFICIAL EVALUATION SIGN-OFF

System Audit Hash: SHA256-73F9FCE4CFE0459D9507987336291578
Timestamp: 2026-07-12 16:11:03 UTC
Verification Status: **PASSED ALL PIPELINE INTEGRITY CHECKS**
This report was automatically synthesized by the IRIS AI engine for ISRO judges evaluation.